

1. Armstrong number

$$\text{Ex}=153 = 1^3 + 5^3 + 3^3 = 153$$

Code:

```
import java.util.Scanner;
```

```
class Armstrong {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter a number: ");  
        int n = sc.nextInt();  
  
        int original = n;  
        int sum = 0;  
  
        for (; n > 0; n = n / 10) {  
            int digit = n % 10;  
            sum = sum + digit * digit * digit;  
        }  
  
        if (sum == original)  
            System.out.println(original + " is an Armstrong  
Number");  
        else  
            System.out.println(original + " is not an  
Armstrong Number");  
    }  
}
```

2. Perfect number

```
import java.util.Scanner;

class PerfectNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;

        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0) {
                sum = sum + i;
            }
        }

        if (sum == n)
            System.out.println(n + " is a Perfect Number");
        else
            System.out.println(n + " is not a Perfect Number");
    }
}
```

3.Palindrome

```
import java.util.Scanner;

class Palindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int n = sc.nextInt();

        int original = n;

        int reverse = 0;

        for (; n > 0; n = n / 10) {

            int digit = n % 10;

            reverse = reverse * 10 + digit;

        }

        if (reverse == original)

            System.out.println(original + " is a Palindrome Number");

        else

            System.out.println(original + " is not a Palindrome Number");

    }

}
```